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SPACER STRUCTURE FOR SEMICONDUCTOR DEVICE

Abstract

The present disclosure describes a semiconductor structure and a method for forming the same. The semiconductor structure can include a substrate, a fin structure over the substrate, a gate structure over the fin structure, a first inner spacer layer formed in the fin structure and adjacent to the gate structure, and a second inner spacer layer extending through the first inner spacer layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. Non-Provisional patent application Ser. No. 18/309,433, filed on Apr. 28, 2023, titled “Spacer Structure for Semiconductor Device,” which is a continuation of U.S. Non-Provisional patent application Ser. No. 17/706,339, filed on Mar. 28, 2022, titled “Spacer Structure for Semiconductor Device,” which is a continuation of U.S. Non-Provisional patent application Ser. No. 16/990,865, filed on Aug. 11, 2020, titled “Spacer Structure for Semiconductor Device,” all of which are incorporated by reference herein in their entireties.

BACKGROUND

[0002] Advances in semiconductor technology has increased the demand for semiconductor devices with higher storage capacity, faster processing systems, higher performance, and lower costs. To meet these demands, the semiconductor industry continues to scale down the dimensions of semiconductor devices, such as nano-sheet FETs. Such scaling down has increased the complexity of semiconductor manufacturing processes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of this disclosure are best understood from the following detailed description when read with the accompanying figures.

[0004] FIG. 1A is an isometric view of a semiconductor device, according to some embodiments.

[0005] FIGS. 1B and 1C are cross-sectional views of a semiconductor device, according to some embodiments.

[0006] FIG. 2 is a flow diagram of a method for fabricating a semiconductor device, according to some embodiments.

[0007] FIG. 3 is an isometric view of a semiconductor device at a stage of its fabrication process, according to some embodiments.

[0008] FIGS. 4A-11A and 4B-11B are cross-sectional views of a semiconductor device at various stages of its fabrication process, according to some embodiments.

[0009] FIG. 12 illustrates a plan view of a semiconductor device manufacturing apparatus, according to some embodiments.

[0010] FIG. 13 illustrates various operating position of a platen of a semiconductor device manufacturing apparatus, according to some embodiments.

[0011] Illustrative embodiments will now be described with reference to the accompanying drawings. In the drawings, like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

[0012] It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “exemplary,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect

such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

[0013] It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

[0014] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0015] The term “nominal” as used herein refers to a desired, or target, value of a characteristic or parameter for a component or a process operation, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The range of values is typically due to slight variations in manufacturing processes or tolerances.

[0016] In some embodiments, the terms “about” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g., +1%, +2%, +3%, +4%, +5% of the value). These values are merely examples and are not intended to be limiting. The terms “about” and “substantially” can refer to a percentage of the values as interpreted by those skilled in relevant art(s) in light of the teachings herein.

[0017] As used herein, the term “vertical” means nominally perpendicular to the surface of a substrate.

[0018] Fins associated with fin field effect transistors (finFETs) or gate-all-around (GAA) FETs can be patterned by any suitable method. For example, the fins can be patterned using one or more photolithography processes, including a double-patterning process or a multi-patterning process. Double-patterning and multi-patterning processes can combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers can then be used to pattern the fins.

[0019] Technology advances in the semiconductor industry drive the pursuit of integrated circuits (ICs) having higher device density, higher performance, and lower cost. In the course of the IC evolution, nano-sheet transistors can replace planar field effect transistor (FET) and/or fin field-effect transistor (finFET) to achieve ICs with higher device densities. Nano-sheet transistors can use a gate-all-around (GAA) gate structure to surround each nano-sheet channel layer to effectively reconcile short channel effects. Nano-sheet transistors require an inner spacer to physically separate the nano-sheet transistor's source-drain (S/D) regions from the nano-sheet transistor's GAA gate structure. However, the inner spacer can be susceptible to structural damage during the fabrication process of defining nano-sheet channel layers. Such inner spacer's structural damage can cause the S/D region's structural damage, thus causing IC failure.

[0020] The present disclosure is directed to a fabrication method and a transistor with a multilayered inner spacer. The transistor can be a gate-all-around field effect transistor (GAA FET), and the multilayered inner spacer can be formed between the transistor's S/D region and the transistor's gate structure. The multilayered inner spacer can have a first dielectric layer and a second dielectric layer formed at sides of the first dielectric layer. Both the first and second dielectric layers can be in contact with the transistor's gate structure. The first dielectric layer can be formed prior to forming the transistor's S/D region, and the second dielectric layer can be formed after forming the transistor's S/D region. The second dielectric layer can protect the S/D

region's structural integrity during the subsequent fabrication process of defining the transistor's nano-sheet channels. A benefit of the present disclosure, among others, is to avoid the S/D region's damage and improve electrical insulation between the S/D region and the gate structure, thus improving the GAA FET's reliability and yield.

[0021] A semiconductor device **100** having multiple FETs **101** formed over a substrate **102** is described with reference to FIGS. **1A-1C**, according to some embodiments. FIG. **1A** illustrates an isometric view of semiconductor device **100**, according to some embodiments. FIG. **1B** illustrates a cross-sectional (e.g., along the x-z plane) view of semiconductor device **100** along line B-B of FIG. **1A**, according to some embodiments. FIG. **1C** illustrates a cross-sectional (e.g., along the x-y plane) view of semiconductor device **100** along line C-C of FIG. **1B**, according to some embodiments. The discussion of elements in FIGS. **1A-1C** with the same annotations applies to each other, unless mentioned otherwise. Semiconductor device **100** can be included in a microprocessor, memory cell, or other integrated circuit (IC). Also, each FET **101** shown in FIGS. **1A-1C** can be a GAA FET, according to some embodiments.

[0022] Referring to FIGS. **1A-1C**, substrate **102** can be a semiconductor material, such as silicon. In some embodiments, substrate **102** can include a crystalline silicon substrate (e.g., wafer). In some embodiments, substrate **102** can include (i) an elementary semiconductor, such as silicon (Si) or germanium (Ge); (ii) a compound semiconductor including silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb); (iii) an alloy semiconductor including silicon germanium carbide (SiGeC), silicon germanium (SiGe), gallium arsenic phosphide (GaAsP), gallium indium phosphide (InGaP), gallium indium arsenide (InGaAs), gallium indium arsenic phosphide (InGaAsP), aluminum indium arsenide (InAlAs), and/or aluminum gallium arsenide (AlGaAs); or (iv) a combination thereof. Further, substrate **102** can be doped depending on design requirements (e.g., p-type substrate or n-type substrate). In some embodiments, substrate **102** can be doped with p-type dopants (e.g., boron (B), indium (In), aluminum (Al), or gallium (Ga)) or n-type dopants (e.g., phosphorus (P) or arsenic (As)).

[0023] FET **101** can include a fin structure **108** extending along an x-direction, a gate structure **110** traversing through fin structure **108** along a y-direction, and S/D regions **124** formed over portions of fin structure **108**. Although FIG. **1A** shows fin structure **108** accommodating two FETs **101**, any number of FETs **101** can be disposed along fin structure **108**. In some embodiments, FET **101** can include multiple fin structures **108** extending along a first horizontal direction (e.g., in the x-direction) and gate structure **110** traversing through the multiple fin structures **108** along a second horizontal direction (e.g., in the y-direction).

[0024] Fin structure **108** can include a buffer region **120** formed over substrate **102**. Fin structure **108** can further include one or more channel regions **122** formed over buffer region **120**. Each channel region **122** can be wrapped by gate structure **110** to function as FET **101**'s channel. For example, a top surface, side surfaces, and a bottom surface of each channel region **122** can be surrounded and in physical contact with gate structure **110**. Buffer region **120** and channel region **122** can be made of materials similar to (e.g., lattice mismatch within 5%) substrate **102**. In some embodiments, each of buffer region **120** and channel region **122** can be made of Si or SiGe. Each of buffer region **120** and channel region **122** can be un-doped, doped with p-type dopants, doped with n-type dopants, or doped with intrinsic dopants. In some embodiments, buffer region **120** and channel regions **122** can be together doped with p-type dopants or together doped with n-type dopants.

[0025] Gate structure **110** can be a multilayered structure that wraps around each channel region **122** to modulate FET **101**. Gate structure **110** can have a length L_{110} representing FET **101**'s channel length. Length L_{110} can have any suitable horizontal (e.g., in the x-direction) dimension, such as from about 5 nm to about 200 nm. Gate structure **110** can include a gate dielectric layer **112** and a gate electrode **114** disposed on dielectric layer **112**. Gate dielectric layer

112 can include any suitable dielectric material with any suitable thickness that can provide channel modulation for FET **101**. In some embodiments, gate dielectric layer **112** can be made of silicon oxide or a high-k dielectric material (e.g., hafnium oxide or aluminum oxide). In some embodiments, gate dielectric layer **112** can have a thickness ranging from about 1 nm to about 5 nm. Based on the disclosure herein, other materials and thicknesses for gate dielectric layer **112** are within the scope and spirit of this disclosure. Gate electrode **114** can function as a gate terminal for FET **101**. Gate electrode **114** can include any suitable conductive material that provides a suitable work function to modulate FET **101**. In some embodiments, gate electrode **114** can be made of titanium nitride, tantalum nitride, tungsten nitride, titanium, aluminum, copper, tungsten, tantalum, copper, or nickel. Based on the disclosure herein, other materials for gate electrode **114** are within the scope and spirit of this disclosure.

[0026] S/D regions **124** can be formed over opposite sides (e.g., along x-direction) of each channel region **122** to function as FET **101**'s source and drain terminals. S/D region **124** can have any suitable lateral (e.g., in the y-direction) width $W_{sub.124}$ such as from about 20 nm to about 200 nm. S/D region **124** can be made of an epitaxially-grown semiconductor material similar to (e.g., lattice mismatch within 5%) channel region **122**. In some embodiments, S/D region **124** can be made of Si, Ge, SiGe, InGaAs, or GaAs. S/D region **124** can be doped with p-type dopants, n-type dopants, or intrinsic dopants. In some embodiments, S/D region **124** can have a different doping type from channel region **122**.

[0027] Semiconductor device **100** can further include a gate spacer **104** formed between gate structure **110** and S/D region **124**. Gate spacer **104** can have a front surface **104F** (shown in FIG. **1C**) formed over and in contact with gate structure **110**. In some embodiments, gate spacer **104** can be further formed over fin structure **108**'s side surface. Gate spacer **104** can be made of any suitable dielectric material. In some embodiments, gate spacer **104** can be made of silicon oxide, silicon nitride, or a low-k material with a dielectric constant less than about 3.9. In some embodiments, gate spacer **104** can have any suitable thickness $t_{sub.104}$, such as from about 5 nm to about 15 nm. Based on the disclosure herein, other materials and thicknesses for gate spacer **104** are within the scope and spirit of this disclosure.

[0028] Semiconductor device **100** can further include shallow trench isolation (STI) regions **138** (shown in FIG. **1A**) configured to provide electrical isolation between fin structures **108**. Also, STI regions **138** can provide electrical isolation between FET **101** and neighboring active and passive elements (not shown in FIG. **1A**) integrated with or deposited on substrate **102**. STI regions **138** can include one or more layers of dielectric material, such as a nitride layer, an oxide layer disposed on the nitride layer, and an insulating layer disposed on the nitride layer. In some embodiments, the insulating layer can include silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), a low-k dielectric material, and/or other suitable insulating materials. Based on the disclosure herein, other dielectric materials for STI region **138** are within the scope and spirit of this disclosure.

[0029] Semiconductor device **100** can further include an interlayer dielectric (ILD) layer **130** (shown in FIGS. **1A** and **1B**) to provide electrical isolation to structural elements it surrounds or covers, such as gate structure **110** and S/D regions **124**. In some embodiments, gate spacer **104** can be formed between gate structure **110** and ILD layer **130**. ILD layer **130** can include any suitable dielectric material to provide electrical insulation, such as silicon oxide, silicon dioxide, silicon oxycarbide, silicon oxynitride, silicon oxy-carbon nitride, and silicon carbonitride. ILD layer **130** can have any suitable thickness, such as from about 50 nm to about 200 nm, to provide electrical insulation. Based on the disclosure herein, other insulating materials and thicknesses for ILD layer **130** are within the scope and spirit of this disclosure.

[0030] Semiconductor device **100** can further include an inner spacer structure **160** formed protruding into fin structure **108**. Inner spacer structure **160** can separate gate structure **110** from S/D region **124**. For example, inner spacer structure **160** can be formed at gate structure **110**'s

opposite sides along FET **101**'s channel direction (e.g., along the x-direction) to separate gate structure **110** from S/D region **124**. In some embodiments, inner spacer structure **160** can be formed between two laterally (e.g., in the y-direction) adjacent gate spacers **104**. In some embodiments, inner spacer structure **160** can be formed between two vertically (e.g., in the z-direction) adjacent channel regions **122**. In some embodiments, inner spacer structure **160** can be formed between buffer region **120** and channel region **122**. In some embodiments, inner spacer structure **160** can be formed between gate structure **110** and ILD layer **130**.

[0031] Referring to FIGS. **1B** and **1C**, inner spacer structure **160** can include a first inner spacer **162** extending from gate structure **110** to S/D region **124**. First inner spacer **162** can be formed in fin structure **108**. For example, first inner spacer **162** can be formed between and in contact with two vertically (e.g., in the z-direction) adjacent channel regions **122**. First inner spacer **162** can be separated from the laterally (e.g., in the y-direction) adjacent gate spacers **104**. For example, first inner spacer **162** can have side surfaces **162S** (shown in FIG. **1C**) separated from the laterally (e.g., in the y-direction) adjacent gate spacers **104**. In some embodiments, first inner spacer **162**'s side surface **162S** can be substantially perpendicular to substrate **102**. In some embodiments, first inner spacer **162**'s side surface **162S** can be substantially parallel with channel region **122**'s side surface. First inner spacer **162** can further have a front surface **162F** (shown in FIG. **1C**) proximate to gate structure **110**. Front surface **162F** can have a first portion **162F1** (shown in FIG. **1C**) proximate to front surface **162F**'s midpoint (e.g., proximate to front surface **162F**'s vertex) and a second portion **162F2** (shown in FIG. **1C**) proximate to side surface **162S**. In some embodiments, front surface **162F** can be a substantially planar surface (not shown in FIG. **1C**), where front surface **162F** can be in contact with gate structure **110**. In some embodiments, front surface **162F** and gate spacer **104**'s front surface **104F** can be substantially coplanar with gate structure **110**. In some embodiments, front surface **162F** can be a curved surface, where front surface **162F**'s first portion **162F1** can be in contact with gate structure **110**, and front surface **162F**'s second portion **162F2** can be separated from gate structure **110**. In some embodiments, first portion **162F1** and gate spacer **104**'s front surface **104F** can be substantially coplanar with gate structure **110**. In some embodiments, gate spacer **104**'s front surface **104F** can be substantially coplanar with gate structure **110**, and first inner spacer **162**'s front surface **162F** (e.g., both first and second portions **162F1** and **162F2**) can protrude into gate structure **110** (not shown in FIGS. **1A-1C**). In some embodiments, first and second portions **162F1** and **162F2** can be substantially coplanar with each other (not shown in FIG. **1C**). In some embodiments, first and second portions **162F1** and **162F2** can be separated from gate structure **110** (not shown in FIG. **1C**). First inner spacer **162** can further have a back surface, horizontally (e.g., in the x-direction) opposite to front surface **162F**, in contact with S/D region **124**. In some embodiments, first inner spacer **162**'s back surface and the laterally (e.g., in the y-direction) adjacent gate spacers **104** can be substantially coplanar with S/D region **124**. First inner spacer **162** can be made of any suitable insulating material to electrically separate gate structure **110** from S/D region **124**. In some embodiments, first inner spacer **162** can be made of amorphous silicon, silicon oxide, silicon nitride, aluminum oxide, hafnium oxide, or a low-k dielectric material. In some embodiments, first inner spacer **162** can be made of dielectric materials free from carbon, chlorine, and/or fluorine. Based on the disclosure herein, other materials for first inner spacer **162** are within the scope and limit of this disclosure.

[0032] Inner spacer structure **160** can further include a second inner spacer **164** formed between first inner spacer **162** and the laterally (e.g., in the y-direction) adjacent gate spacers **104**. For example, second inner spacer **164** can be formed over first inner spacer **162**'s side surfaces **162S** and in contact with laterally (e.g., in the y-direction) adjacent gate spacers **104**. Second inner spacer **164** can have a front surface **164F** (shown in FIG. **1C**) in contact with gate structure **110**. In some embodiments, second inner spacer **164** can be further formed over front surface **162F**'s second portion **162F2**, where second inner spacer **164**'s front surface **164F** can be positioned between gate structure **110** and front surface **162F**'s second portion **162F2**. In some embodiments, front surface

162F's first portion **162F1** and second inner spacer **164**'s front surface **164F** can be substantially coplanar with gate structure **110**. In some embodiments, gate spacer **104**'s front surface **104F** can be substantially coplanar with gate structure **110**, and second inner spacer **164**'s front surface **164F** can protrude into gate structure **110** (not shown in FIGS. **1A-1C**). Second inner spacer **164** can be made of any suitable insulating material to electrically separate gate structure **110** from S/D region **124**. In some embodiments, second inner spacer **164** can be made of an oxide material, such as silicon oxide and silicon oxynitride (SiON). In some embodiments, second inner spacer **164** can be made of an carbon-contained oxide material, such as silicon oxycarbide (SiOC) and silicon oxycarbonitride (SiNOC). In some embodiments, second inner spacer **164** can be made of a polymer material. In some embodiments, second inner spacer **164** can be made of an oxide material that can include carbon, chlorine, and/or fluorine (discussed in method **200**), and first inner spacer **162** and/or third inner spacer **106** (discussed below) can be made of dielectric materials free from carbon, chlorine, and/or fluorine. In some embodiments, first and second inner spacers **164** and **162** can be made of different material from each other.

[0033] Inner spacer structure **160** can further include a third inner spacer **106** (shown in FIG. **1C**) formed between first inner spacer **162** and the laterally (e.g., in the y-direction) adjacent gate spacers **104**. For example, third inner spacer **106** can be formed over first inner spacer **162**'s side surfaces **162S** and in contact with the laterally (e.g., in the y-direction) adjacent gate spacers **104**. Third inner spacer **106** can be further formed proximate to S/D region **124** and separated from gate structure **110**. For example, third inner spacer **106** can be formed over a first portion of first inner spacer **162**'s side surfaces **162S** that is proximate to S/D region **124**, and second inner spacer **164** can be formed over a second portion of first inner spacer **162**'s side surfaces **162S** that is proximate to gate structure **110**. In some embodiments, third inner spacer **106** and gate spacer **104** can be substantially coplanar with S/D region **124**. Third inner spacer **106** can be made of any suitable insulating material to electrically separate gate structure **110** from S/D region **124**. In some embodiments, third inner spacer **106** can be made of an oxide material, such as silicon oxide. In some embodiments, third inner spacer **106** can be made of dielectric materials free from carbon, chlorine, and/or fluorine. In some embodiments, third inner spacer **106** and second inner **162** spacer can be made of oxide materials that can have different oxide concentrations from each other. In some embodiments, first, second, and third inner spacers **164**, **162**, and **106** can be made of different materials from one another. Based on the disclosure herein, other materials for third inner spacer **106** are within the scope and limit of this disclosure.

[0034] FIG. **2** is a flow diagram of a method **200** for fabricating semiconductor device **100**, according to some embodiments. For illustrative purposes, the operations illustrated in FIG. **2** will be described with reference to the example fabrication process for fabricating semiconductor device **100** as illustrated in FIGS. **3**, **4A-11A**, **4B-11B**, and **12**. FIG. **3** illustrates an isometric view of semiconductor device **100** at a stage of its fabrication, according to some embodiments. FIGS. **4A-11A** and **4B-11B** illustrate cross-sectional views along lines B-B and C-C, respectively, of structure of FIG. **3** at various stages of its fabrication, according to some embodiments. FIG. **12** illustrates an exemplary apparatus for forming inner spacer structure **160** in method **200**, according to some embodiments. FIG. **13** illustrates various operating positions of a platen of the exemplary apparatus for forming inner spacer structure **160** in method **200**, according to some embodiments. Operations can be performed in a different order or not performed depending on specific applications. Method **200** may not produce a complete semiconductor device **100**. Accordingly, it is understood that additional processes can be provided before, during, and/or after method **200**, and that some other processes may be briefly described herein. Further, the discussion of elements in FIGS. **1A-1C**, **3**, **4A-11A**, **4B-11B**, and **12** with the same annotations applies to each other, unless mentioned otherwise.

[0035] Referring to FIG. **2**, in operation **205**, a recess structure is formed in a fin structure. For example, a recess structure **436** (shown in FIGS. **4A** and **4B**) can be formed in fin structure **108**

(shown in FIG. 4A) with reference to FIGS. 3, 4A, and 4B. The process of forming recess structure **436** can include (i) forming fin structures **108** (shown in FIG. 3) over substrate **102**; (ii) forming STI region **138** (shown in FIG. 3) over the etched substrate **102** using a deposition process and an etch back process; (iii) forming sacrificial gate structures **310** (shown in FIG. 3) with length $L_{\text{sub.110}}$ over fin structures **108**; and (iv) removing fin structures **108** through sacrificial gate structures **310** to form recess structure **436**.

[0036] Referring to FIG. 3, the process of forming fin structures **108** can include (i) providing substrate **102**; (ii) epitaxially growing channel regions **122** and sacrificial layers **322** over substrate **102**; and (iii) etching channel regions **122**, sacrificial layers **322**, and substrate **102** through a patterned mask layer (not shown in FIG. 3) using an etching process.

[0037] Sacrificial layer **322** can be made of materials different from channel region **122** and similar to (e.g., lattice mismatch within 5%) substrate **102**. In some embodiments, sacrificial layer **322** can be made of SiGe, and channel region **122** can be made of Si. In some embodiments, sacrificial layer **322** and channel region **122** can be made of SiGe with different atomic percentage of Ge from each other. Channel region **122** and sacrificial layer **322** can have suitable thicknesses $t_{\text{sub.122}}$ and $t_{\text{sub.322}}$, respectively. In some embodiments, each of thicknesses $t_{\text{sub.122}}$ and $t_{\text{sub.322}}$ can be from about 5 nm to about 10 nm. Channel region **122** and sacrificial layer **322** can be epitaxially grown using any suitable epitaxial growth process, such as a chemical vapor deposition (CVD) process, a low pressure CVD (LPCVD) process, a rapid thermal CVD (RTCVD) process, a metal-organic CVD (MOCVD) process, an atomic layer CVD (ALCVD) process, an ultrahigh vacuum CVD (UHVCVD) process, a reduced pressure CVD (RPCVD) process, a molecular beam epitaxy (MBE) process, a cyclic deposition-etch (CDE) process, and a selective epitaxial growth (SEG) process. Based on the disclosure herein, other materials, thicknesses, and epitaxial growth processes for channel region **122** and sacrificial layer **322** are within the scope and limit of this disclosure.

[0038] The etching process for removing channel region **122**, sacrificial layer **322** and substrate **102** can include a dry etching process or a wet etching process to define fin structure **108** with any suitable width $W_{\text{sub.108}}$, such as from about 5 nm to about 50 nm. In some embodiments, the dry etching process can include using any suitable etchant, such as an oxygen-containing gas, a fluorine-containing gas, a chlorine-containing gas, and a bromine-containing gas, and the wet etching process can include etching in any suitable wet etchant, such as diluted hydrofluoric acid, potassium hydroxide solution, ammonia, and nitric acid. Based on the disclosure herein, other widths and etching processes for fin structure **108** are within the scope and limit of this disclosure.

[0039] The deposition process for forming STI region **138** can include any suitable growth process, such as a physical vapor deposition (PVD) process, a CVD process, a high-density-plasma (HDP) CVD process, a flowable CVD (FCVD) process, and an atomic layer deposition (ALD) process. The etch back process for forming STI region **138** can include a dry etching process, a wet etching process, or a polishing process, such as chemical vapor deposition (CMP) process. Based on the disclosure herein, other processes for forming STI region **138** are within the spirit and scope of this disclosure.

[0040] The process of forming sacrificial gate structure **310** can include (i) blanket depositing a dielectric layer **306** with a suitable thickness $t_{\text{sub.106}}$, such as from about 1 nm to about 5 nm, over fin structures **108** using a suitable deposition process, such as a CVD process, a PVD process, and an ALD process; (ii) blanket depositing a polysilicon layer **348** and a hard mask layer **350** over dielectric layer **306** using a suitable deposition process, such as a CVD process, a PVD process, and an ALD process; (iii) removing dielectric layer **306**, polysilicon layer **348** and hard mask layer **350** through a patterned mask layer (not shown in FIG. 3) using an etching process; and (iv) forming gate spacers **104** with a suitable thickness $t_{\text{sub.104}}$, such as from about 5 nm to about 15 nm, over side surfaces of polysilicon layer **348** and/or over fin structure **108**'s side surfaces using a suitable deposition process and an etching process. In some embodiments, the deposition process

for dielectric layer **306** can use any suitable chlorine-free processing gases and/or fluorine-free processing gases. In some embodiments, dielectric layer **306** can be made of a same material as third inner spacer **106**. Based on the disclosure herein, other processes for forming gate structures **310** are within the spirit and scope of this disclosure.

[0041] Referring to FIGS. **4A** and **4B**, after forming sacrificial gate structure **310**, recess structure **436** can be formed by removing channel regions **122**, sacrificial layers **322**, and substrate **102** through sacrificial gate structures **310** and gate spacers **104** using an etching process. The etching process can include a dry etching process or a wet etching process. In some embodiments, the etching process can be a time-etching process. In some embodiments, the dry etching process can include using any suitable etchant, such as an oxygen-containing gas, a fluorine-containing gas, a chlorine-containing gas, and a bromine-containing gas, and the wet etching process can include etching in any suitable wet etchant, such as diluted hydrofluoric acid, potassium hydroxide solution, ammonia, and nitric acid. As shown in FIG. **4A**, the resulting recess structure **436** can expose fin structure **108**'s side surface **322S**. In some embodiments, side surface **322S** can represent side surfaces of sacrificial layers **322** that are under polysilicon layer **348** and gate spacers **104**. In some embodiments, side surface **322S** can represent side surfaces of channel regions **122** that are under polysilicon layer **348** and gate spacers **104**. Further, as shown in FIG. **4B**, the resulting recess structure **436** can expose gate spacers **104** that are laterally (e.g., in the y-direction) adjacent to dielectric layers **306** and sacrificial layers **322**. Accordingly, gate spacers **104** can have front surface **104F** in contact with polysilicon layer **348** and a back surface, opposite to front surface **104F**, exposed by recess structure **436**.

[0042] Referring to FIG. **2**, in operation **210**, a first inner spacer structure is formed in the fin structure. For example, first inner spacers **162** (shown in FIGS. **5A** and **5B**) can be formed in fin structure **108** of FIGS. **4A** and **4B**. The process of forming first inner spacer **162** can include (i) forming recess structures (not shown in FIGS. **5A** and **5B**) in sacrificial layers **322** using a selective etching process that can selectively etch sacrificial layer **322** from channel region **122**; (ii) blanket depositing first inner spacer **162** in recess structure **436** and in the recess structures in sacrificial layers **322** using a deposition process, such as a CVD process, a PVD process, and an ALD process; and (iii) removing first inner spacer **162** through polysilicon layer **348** and gate spacers **104** using an etching process to define first inner spacer **162** with any suitable thickness $t_{\text{sub.162}}$, such as from about 3 nm to about 10 nm, protruding into sacrificial layers **322**. In some embodiments, first inner spacer **162**'s thickness $t_{\text{sub.162}}$ can be substantially equal to or greater than gate spacer **104**'s thickness $t_{\text{sub.104}}$. In some embodiments, the deposition process for first inner spacer **162** can use any suitable chlorine-free processing gases and/or fluorine-free processing gases. The resulting first inner spacer **162** can have front surface **162F** (e.g., first and second portions **162F1** and **162F2** as shown in FIG. **5B**) protruding into and in contact with sacrificial layer **322**. The resulting first inner spacer **162** can further have side surface **162S** in contact with laterally (e.g., in the y-direction) adjacent dielectric layers **306**. In some embodiments, the resulting first inner spacer **162**'s back surface, opposite to front surface **162F**, can be substantially coplanar with dielectric layer **306** and laterally (e.g., in the y-direction) adjacent gate spacers **104**. Based on the disclosure herein, other thicknesses and other etching processes for forming first inner spacer **162** are within the spirit and scope of this disclosure.

[0043] Referring to FIG. **2**, in operation **215**, an epitaxial region is formed in the recess structure and over the first inner spacer structure. For example, S/D region **124** (shown in FIGS. **6A** and **6B**) with lateral (e.g., in the y-direction) width $W_{\text{sub.124}}$ can be formed in recess structures **436** and over first inner spacers **162** of FIGS. **5A** and **5B**. The process of forming S/D region **124** can include epitaxially growing S/D region **124** in the structure of FIGS. **5A** and **5B** using an epitaxial growth process, such as a CVD process, a LPCVD process, a RTCVD process, a MOCVD process, an ALCVD process, an UHVCVD process, a RPCVD process, an MBE process, a CDE process, and an SEG process. The epitaxial growth process can be performed using suitable precursors, such

as silane (SiH.sub.4), disilane (Si.sub.2H.sub.6), dichlorosilane (DCS), and germane (GeH.sub.4). The epitaxial growth process can further include doping S/D region **124** using suitable dopant precursors, such as diborane (B.sub.2H.sub.6), boron trifluoride (BF.sub.3), phosphine (PH.sub.3), and arsine (AsH.sub.3). In some embodiments, the epitaxial growth process for S/D region **124** can further include applying an etching gas, such as hydrogen chloride (HCl), to adjust S/D **124**'s crystalline facets. Accordingly, the resulting S/D region **124** can be grown over and in contact with channel regions **122** under polysilicon layer **348** and gate spacers **104**. The resulting S/D region **124** can be further grown over and in contact with first inner spacers **162** that are vertically (e.g., in the z-direction) sandwiched by channel regions **122** and laterally (e.g., in the y-direction) sandwiched by gate spacers **104** and dielectric layers **306**. In some embodiments, as shown in FIG. **6A**, portions of first inner spacer **162** that is adjacent to topmost sacrificial layer **322** can be exposed after forming S/D region **124**. Based on the disclosure herein, other epitaxial growth processes for forming S/D region **124** are within the spirit and scope of this disclosure.

[0044] Referring to FIG. **2**, in operation **220**, a second inner spacer structure is formed over the epitaxial region and adjacent to the first inner spacer structure. For example, third inner spacer **106** (shown in FIG. **8B**) and second inner spacer **164** (shown in FIGS. **11A** and **11B**) can be formed over S/D region **124** and adjacent to first inner spacer **162** with reference to FIGS. **7A-11A**, **7B-11B** and **12**. Referring to FIGS. **7A** and **7B**, operation **220** can begin with forming third inner spacer **106** laterally (e.g., in the y-direction) adjacent to first inner spacer **162**. The process of forming third inner spacer **106** can include (i) forming ILD layer **130** coplanarized with sacrificial gate structures **310** of FIGS. **6A** and **6B** using a suitable deposition process, such as a PVD process and a CVD process, and a suitable etch back process, such as a chemical mechanical polishing (CMP) process; (ii) removing hard mask layer **350** and polysilicon layer **348** to form recess structures **736** to expose dielectric layer **306** using an etching process; and (iii) removing portions of dielectric layer **306** to expose sacrificial layers **322** and define third inner spacer **106** over first inner spacer **162**'s side surface **162S** using an etching process. In some embodiments, the etching process for forming recess structure **736** can include a dry etching process that uses chlorine, fluorine or bromine as gas etchants. In some embodiments, the etching process for forming recess structure **736** can include a wet etching process that uses an ammonium hydroxide (NH.sub.4OH), sodium hydroxide (NaOH), or potassium hydroxide (KOH) as wet etchants. In some embodiments, the etching process for removing dielectric layer **306** can include a dry etching process that uses chlorine, fluorine or bromine as gas etchants. In some embodiments, the etching process for removing dielectric layer **306** can include a wet etching process that uses an hydrogen fluoride (HF) as wet etchants.

[0045] Subsequently, operation **220** can further include forming second inner spacer **164** (shown in FIGS. **11A** and **11B**) using an etching-deposition process with reference to FIGS. **8A-11A**, **8B-11B** and **12**. In some embodiments, the etching-deposition process can include a radical deposition process and a radical etching process. The radical deposition process and the radical etching process can avoid channel regions **122**'s crystalline damages after forming second inner spacer **164**, thus being essential for operation **220**. In some embodiments, the radical deposition process and the radical etching process for forming second inner spacer **164** (e.g., including the process for thinning sacrificial layers **322**, the process for depositing dielectric layer **964**, and the process for removing dielectric layer **964**; discussed below) can be performed in a same semiconductor device manufacturing apparatus (e.g., without break vacuum between the radical deposition process and the radical etching process).

[0046] Referring to FIGS. **12** and **13**, a semiconductor device manufacturing apparatus **1200** can be configured to perform the radical deposition process and the radical etching process.

Semiconductor device manufacturing apparatus **1200** can include a chamber **1201**, a platen **1207** housed in chamber **1201** and configured to hold semiconductor device **100**, and a radical generator **1211** disposed over platen **1207**. Platen **1207** can be moved, via a motion mechanism (not shown in

FIG. 12), in chamber 1201. For example, platen 1207 can be moved to position Z.sub.near that is at a distance S.sub.near away from radical generator 1211. Platen 1207 can also be moved to position Z.sub.far that is at a distance S.sub.far, greater than distance S.sub.near, away from radical generator 1211. Radical generator 1211 can receive a processing gas through a gas input (not shown in FIG. 12). Radical generator 1211 can further receive a radio frequency (RF) discharging power from a RF source (not shown in FIG. 12) to convert the processing gas to supply radicals 1202 propagating towards platen 1207. Semiconductor device manufacturing apparatus 1200 can perform the radical deposition process and/or the radical etching process using radicals 1202. In some embodiments, radicals 1202 can perform both the radical etching process and the radical deposition process, where the radical deposition process's deposition rate and the radical etching process's etching rate can be determined based on platen 1207's position relative to radical generator 1211. For example, because radicals 1202 at position Z.sub.near can have a sufficient kinetic energy, the radical etching process, using radicals 1202, at position Z.sub.near can have an etching rate greater than a deposition rate of the radical deposition process, using radicals 1202, at position Z.sub.near. Accordingly, as shown in FIG. 13's radical etching stage portion, platen 1207, which holds semiconductor device 100, can be moved to position Z.sub.near (e.g., with distance S.sub.near away from radical generator 1211) for a suitable time duration t.sub.1, such as from about 3 seconds to about 30 seconds, to perform the radical etching process on semiconductor device 100 using the radicals that have the sufficient kinetic energy. Similarly, because radicals 1202 at position Z.sub.far can have a reduced kinetic energy, the radical deposition process, using radicals 1202, at position Z.sub.far can have a deposition rate greater than an etching rate of the radical etching process, using radicals 1202, at position Z.sub.far. Accordingly, as shown in FIG. 13's radical deposition stage portion, platen 1207, which holds semiconductor device 100, can be moved to position Z.sub.far (e.g., with distance S.sub.far away from radical generator 1211) for a suitable time duration t.sub.2, such as from about 3 seconds to about 30 seconds, to perform the radical deposition process on semiconductor device 100 using the radicals that have the reduced kinetic energy. In some embodiments, distance S.sub.near can be from about 5 mm to about 50 mm or from about 10 mm to about 30 mm for semiconductor device manufacturing apparatus 1200 to perform the radical etching process on semiconductor device 100 at position Z.sub.near. If distance S.sub.near is below the above-noted lower limits, radicals 1202 may have excess kinetic energy to damage semiconductor device 100's structural integrity. If distance S.sub.near is beyond the above-noted upper limits, semiconductor device manufacturing apparatus 1200 may not perform the radical etching process, because radicals 1202 may result in another radical deposition process with a deposition rate greater than the radical etching process's etching rate. In some embodiments, distance S.sub.far can be from about 60 mm to about 120 mm or from about 90 mm to about 100 mm for semiconductor device manufacturing apparatus 1200 to perform the radical deposition process on semiconductor device 100 at position Z.sub.far. If distance S.sub.far is below the above-noted lower limits, semiconductor device manufacturing apparatus 1200 may not perform the radical deposition process because radicals 1202 may result in another radical etching process with an etching rate greater than the radical deposition process's deposition rate. If distance S.sub.near is beyond the above-noted upper limits, radicals 1202 may be susceptible to insufficient kinetic energy to initiate the radical deposition process on semiconductor device 100. In some embodiments, a ratio of distance S.sub.far to distance S.sub.near can be from about 2.0 to about 12.0 or from about 3.0 to about 10.0 to enable semiconductor device manufacturing apparatus 1200 to perform the radical etching process on semiconductor device 100 at position Z.sub.near and the radical deposition process on semiconductor device 100 at position Z.sub.far. If the ratio of distance S.sub.far to distance S.sub.near is below the above-noted lower limits, radicals 1202's kinetic energy may be substantially unchanged between positions S.sub.far and S.sub.near, thus being unable to provide both radical deposition process and radical etching process in semiconductor manufacturing apparatus 1200. If the ratio of distance S.sub.far to distance

S.sub.near is below the above-noted lower limits, radicals **1202** may be susceptible to insufficient kinetic energy to initiate the radical deposition process on semiconductor device **100**. In some embodiments, during the time duration $t_{\text{sub.1}}$ of FIG. **13**, platen **1207**'s position can be vertically (e.g., in the z-direction) adjusted from position Z.sub.near to another position proximate to Z.sub.near (e.g., within position Z.sub.near by about 30% of separation S.sub.near) to perform the radical etching process. In some embodiments, during the time duration $t_{\text{sub.2}}$ of FIG. **13**, platen **1207**'s position can be vertically (e.g., in the z-direction) adjusted from position Z.sub.far to another position proximate to Z.sub.far (e.g., within position Z.sub.far by about 30% of separation S.sub.far) to perform the radical deposition process. In some embodiments, the processing gases, received by radical generator **1211**, for the radical etching process (e.g., during the time duration $t_{\text{sub.1}}$ of FIG. **13**) and the radical deposition process (e.g., during the time duration $t_{\text{sub.2}}$ of FIG. **13**) can be made of different gas species. In some embodiments, the processing gases, received by radical generator **1211**, for the radical etching process (e.g., during the time duration $t_{\text{sub.1}}$ of FIG. **13**) and the radical deposition process (e.g., during the time duration $t_{\text{sub.2}}$ of FIG. **13**) can be made of identical gas species. In some embodiments, radical generator **1211** can receive the processing gases to generate radicals **1202** while platen **1207** is moving between position Z.sub.near and position Z.sub.far (e.g., switching between the radical deposition process and the radical etching process).

[0047] Referring to FIGS. **8A** and **8B**, the process of forming second inner spacer **164** can include thinning sacrificial layer **322** of FIGS. **7A** and **7B**. In some embodiments, the process of thinning sacrificial layer **322** can be a plasma etching process. In some embodiments, the process of thinning sacrificial layer **322** can be the radical etching process. For example, semiconductor device **100** of FIGS. **7A** and **7B** can be placed on platen **1207** (shown in FIG. **12**), and platen **1207** can be subsequently moved to position Z.sub.near (shown in FIG. **12**) to perform the radical etching process for time duration $t_{\text{sub.1}}$ (shown in FIG. **13**) at or proximate to position Z.sub.near. The processing gas associated with the radical etching process for thinning sacrificial layer **322** can include a fluorine-based gas, such as nitrogen trifluoride ($\text{NF}_{\text{sub.3}}$) and carbon tetrafluoride ($\text{CF}_{\text{sub.4}}$). In some embodiments, the processing gas associated with the radical etching process can include a chlorine-based gas or a bromine-based gas. Accordingly, as shown in FIG. **8A**, sacrificial layer **322** can be reduced from thickness $t_{\text{sub.322}}$ (e.g., about 10 nm) to a thickness $t_{\text{sub.822}}$ (e.g., about 5 nm), less than thickness $t_{\text{sub.322}}$, to form recess structures **801**, vertically (e.g., in the z-direction) adjacent to sacrificial layer **322**, to expose vertically (e.g., in the z-direction) adjacent channel regions **122**. Further, as shown in FIG. **8B**, sacrificial layer **322** can be reduced from width $W_{\text{sub.108}}$ (e.g., about 20 nm) to a width $W_{\text{sub.822}}$ (e.g., about 10 nm), less than width $W_{\text{sub.108}}$, to form recess structures **801**, laterally (e.g., in the y-direction; shown in FIG. **8B**) adjacent to sacrificial layer **322**, to expose first inner spacer **162**'s side surfaces **162S** and first inner spacer **162**'s second portion **162F2**. In some embodiments, the radical etching process can further remove portions of third inner spacer **106** to form recess structure **803**. Accordingly, recess structure **803** can protrude into third inner spacer **106** to define third inner spacer **106**'s front surface **106F** that separates from S/D region **124** with a separation $S_{\text{sub.106}}$. In some embodiments, separation $S_{\text{sub.106}}$ can represent the horizontal (e.g., in the x-direction) dimension of FIG. **1C**'s third inner spacer **106** after method **200**. In some embodiments, because the radical etching process can have a reduced RF power compared to plasma processes (e.g., a reactive ion etching process), recess structure **803** does not extend through third inner spacer **106**. Accordingly, separation $S_{\text{sub.106}}$ can be greater than zero, such as from greater than about 1 nm and greater about 2 nm, to allow third inner spacer **106** to protect separate S/D region **124** from the radical etching process during the process of forming recess structures **801** and/or **803**. If separation $S_{\text{sub.106}}$ is below the above-noted lower limits, the straggle of the radical etching process for forming recess structures **801** and/or **803** may penetrate through third inner spacer **106** to damage S/D region **124**'s structural integrity. In some embodiments, separation $S_{\text{sub.106}}$ can be greater

than zero and less than gate spacer **104**'s thickness $t_{\text{sub.104}}$.

[0048] Referring to FIGS. **9A** and **9B**, the process of forming second inner spacer **164** can further include depositing a dielectric layer **964** over the structure of FIGS. **8A** and **8B** using the radical deposition process. For example, semiconductor device **100** of FIGS. **8A** and **8B** can be placed on platen **1207** (shown in FIG. **12**), and platen **1207** can be moved away from radical generator **1211** (shown in FIG. **12**), such as moving from position Z.sub.near to position Z.sub.far (shown in FIG. **12**), to perform the radical deposition process for time duration $t_{\text{sub.2}}$ (shown in FIG. **13**) at or proximate to position Z.sub.far. In some embodiments, the process of thinning sacrificial layers **322** and the process of depositing dielectric layer **964** can be performed in semiconductor device manufacturing apparatus **1200** by moving platen **1207** towards and away from, respectively, radical generator **1211** without breaking chamber **1201**'s vacuum. The processing gas associated with the radical deposition process for depositing a dielectric layer **964** can include oxygen, a fluorine-based gas (e.g., CF_4 or NF_3), a silicon-based gas, such as SiH_4 , or a chlorine-based gas, such as silicon tetrachloride (SiCl_4). Accordingly, dielectric layer **964** can be made of a same material as second inner spacer **164** to fill recess structure **801** to cover channel regions **122** and sacrificial layers **322**. For example, dielectric layer **964** can be made of a dielectric material that can include carbon, chlorine, and/or fluorine. Further, dielectric layer **964** can fill recess structure **803** to cover third inner spacer **106**, first inner spacer **162**'s side surface **162S**, and first inner spacer **162**'s second portion **162F2**. In some embodiments, the processing gas associated with the radical deposition process for depositing dielectric layer **964** can be different from another processing gas associated with the radical etching process for thinning sacrificial layer **322**. In some embodiments, the processing gas associated with the radical deposition process for depositing dielectric layer **964** can be the same as another processing gas associated with the radical etching process for thinning sacrificial layer **322**.

[0049] Referring to FIGS. **10A** and **10B**, the process of forming second inner spacer **164** can include removing dielectric layer **964** to expose gate spacers **104** and channel regions **122** using the radical etching process. For example, semiconductor device **100** of FIGS. **9A** and **9B** can be placed on platen **1207** (shown in FIG. **12**), and platen **1207** can be moved towards radical generator **1211** (shown in FIG. **12**), such as moving from position Z.sub.far to position Z.sub.near (shown in FIG. **12**), to perform the radical etching process for time duration $t_{\text{sub.1}}$ (shown in FIG. **13**) at or proximate to position Z.sub.near. Because the radical etching process for removing dielectric layer **964** can have a reduced RF power compared to plasma processes (e.g., a reactive ion etching process), crystalline damage of the exposed channel region **122** can be avoided after forming second inner spacers **164**. In some embodiments, the processing gas associated with the radical etching process for removing dielectric layer **964** can include a fluorine-based gas, such as NF_3 and CF_4 . In some embodiments, the processing gas associated with the radical etching process can include a chlorine-based gas or a bromine-based gas. In some embodiments, the processing gas associated with the radical etching process for removing dielectric layer **964** can be the same as another processing gas associated with the radical etching process for thinning sacrificial layer **322**. The radical etching process for removing dielectric layer **964** can etch a first portion of dielectric layer **964** to expose gate spacers **104**, while a second portion of dielectric layer **964** can remain filled in recess structure **803** (shown in FIG. **9B**), after the radical etching process. Such second portion of dielectric layer **964** filled in recess structure **803** can therefore define second inner spacer **164** (shown in FIG. **10B**) to protect third inner spacer **106** and S/D region **124** from structural damages introduced by the radical etching process.

[0050] Further, the radical etching process for removing dielectric layer **964** can thin sacrificial layers **322** (e.g., thinning sacrificial layer **322** from thickness $t_{\text{sub.822}}$ (e.g., about 5 nm) to a thickness $t_{\text{sub.1022}}$ (e.g., about 3 nm) less than thickness $t_{\text{sub.822}}$; thinning sacrificial layer **322** from width $W_{\text{sub.822}}$ (e.g., about 10 nm) to a width $W_{\text{sub.1022}}$ (e.g., about 5 nm) less than width $W_{\text{sub.822}}$). In some embodiments, with sacrificial layers **322** remaining (e.g., thickness $t_{\text{sub.1022}}$

and width $W_{\text{sub.1022}}$ greater than zero) after removing dielectric layer **964**, the process of forming second inner spacer **164** can loop back to (i) deposit another dielectric layer **964** using the radical deposition process (e.g., discussed previously at FIGS. **9A** and **9B**) and (ii) remove the other dielectric layer **964** using the radical deposition process (e.g., discussed previously at FIGS. **10A** and **10B**) to further thin sacrificial layers **322**. In some embodiments, referring to FIGS. **11A** and **11B**, the radical etching process for removing dielectric layer **964** can completely remove sacrificial layers **322** to form recess structures **1101** connecting two vertically (e.g., in the z-direction) adjacent channel regions **122**. Accordingly, the resulting second inner spacer **164** with a horizontal (e.g., in the x-direction) thickness $t_{\text{sub.164}}$ can be formed over first inner spacer **162**'s side surfaces **162S** and second portion **162F2**, and first inner spacer **162**'s first portion **162F1**, previously connecting with sacrificial layer **322**, can be exposed to recess structure **1101**. In some embodiments, thickness $t_{\text{sub.164}}$ can represent the horizontal (e.g., in the x-direction) dimension of FIG. **1C**'s second inner spacer **164** after method **200**. Thickness $t_{\text{sub.164}}$ can be from about 2 nm to about 5 nm or from about 2 nm to about 3 nm. If thickness $t_{\text{sub.164}}$ is below the above-noted lower limits, second inner spacer **164** may not be able to protect third inner spacer **106**'s structural integrity from the radical etching process's straggle during the process of removing dielectric layer **964**. If thickness $t_{\text{sub.164}}$ is beyond the above-noted upper limits, second inner spacer **164** may reduce the contact area between channel region **122** and gate structure **110** (formed at operation **225**; discussed below), thus degrading gate structure **110**'s gate modulation to reconcile FET **101**'s short channel effect. In some embodiments, a ratio of thickness $t_{\text{sub.162}}$ to thickness $t_{\text{sub.164}}$ can be from about 0.2 to about 0.8, from about 0.4 to about 0.8, or from about 0.4 to about 0.6. If the ratio of thickness $t_{\text{sub.162}}$ to thickness $t_{\text{sub.164}}$ is below the above-noted lower limits, second inner spacer **164** may reduce the contact area between channel region **122** and gate structure **110** (formed at operation **225**; discussed below), thus degrading gate structure **110**'s gate modulation to reconcile FET **101**'s short channel effect. If the ratio of thickness $t_{\text{sub.162}}$ to thickness $t_{\text{sub.164}}$ is beyond the above-noted upper limits, second inner spacer **164** may not be able to protect third inner spacer **106**'s structural integrity from the radical etching process's straggle during the process of removing dielectric layer **964**. In some embodiments, thickness $t_{\text{sub.164}}$ can be substantially equal to the difference between separation $S_{\text{sub.106}}$ and thickness $t_{\text{sub.104}}$ (e.g., front surfaces **104F** and **164F** can be substantially coplanar with each other). In some embodiments, the sum of thickness $t_{\text{sub.164}}$ and separation $S_{\text{sub.106}}$ can be substantially equal to thickness $t_{\text{sub.162}}$.

[0051] Referring to FIG. **2**, in operation **225**, a metal gate structure is formed over the fin structure. For example, gate structure **110** (shown in FIGS. **1A** and **1B**) can be formed over fin structure **108**. The process of forming gate structure **110** can include filling gate dielectric layer **112** and a gate electrode **114** in the recess structures **1101** of FIGS. **11A** and **11B** using a suitable deposition process, such as an ALD process and a CVD process, and a suitable etch back process, such as a CMP process. Based on the disclosure herein, other processes for forming gate structure **110** are within the spirit and scope of this disclosure.

[0052] The present disclosure provides an exemplary transistor inner spacer structure and a method for forming the same. The inner spacer structure can have a multilayered structure to improve its structural integrity to protect the transistor's S/D structure. The inner spacer structure can have a first dielectric layer connecting to the transistor's S/D structure and the transistor's gate structure. The inner spacer structure can further have second and third dielectric layers formed over first dielectric layer's side surfaces. The second dielectric layer can be proximate to the transistor's S/D structure, and the third dielectric layer can be proximate to the transistor's gate structure. The method of forming the inner spacer structure can include performing an etching-deposition process in a semiconductor device manufacturing apparatus. The etching-deposition process can include a radical deposition process to deposit the third dielectric layer. The etching-depositing process can further include a radical etching process to remove the sacrificial semiconductor layers to define

the transistor's channel. The radical etching process and the radical deposition process can be performed in the same semiconductor device manufacturing apparatus without breaking the vacuum of in the semiconductor device manufacturing apparatus. A benefit of the present disclosure, among others, is to provide the inner structure with robust structural integrity and the in-situ etching-deposition process for forming the same, thus improving the IC's reliability and throughput.

[0053] In some embodiments, a semiconductor structure can include a substrate, a fin structure over the substrate, a gate structure over the fin structure, a first inner spacer layer formed in the fin structure and adjacent to the gate structure, and a second inner spacer layer extending through the first inner spacer layer.

[0054] In some embodiments, a semiconductor structure can include a substrate, a fin structure over the substrate, a gate structure formed in the fin structure, a source/drain (S/D) region formed in the fin structure and separated from the gate structure, a first inner spacer layer formed in the fin structure and adjacent to the gate structure, and a second inner spacer layer extending from the gate structure to the S/D region. The first inner spacer layer can be over a side surface of the second inner spacer layer.

[0055] In some embodiments, a method can include forming a fin structure over a substrate, forming a first recess structure in the fin structure to expose a side surface of the fin structure, forming a first inner spacer layer protruding into the side surface of the fin structure, forming a second recess structure in the fin structure to expose an other side surface of the first inner spacer layer, and forming a second inner spacer layer over the other side surface.

[0056] The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A structure, comprising: channel regions over a substrate; a gate structure surrounding the channel regions; a first inner spacer between the channel regions and having a front surface in contact with the gate structure; and a second inner spacer in contact with the gate structure and the front surface of the first inner spacer.
2. The structure of claim 1, further comprising a third inner spacer in contact with a side surface of the first inner spacer and a back surface of the second inner spacer, wherein the side surface is perpendicular to the back surface.
3. The structure of claim 1, wherein the first inner spacer comprises a dielectric material free from carbon.
4. The structure of claim 1, wherein the second inner spacer comprises a carbon-contained oxide material.
5. The structure of claim 1, wherein the second inner spacer is in contact with a side surface of the first inner spacer.
6. The structure of claim 1, wherein an interface between the first and second inner spacers is curved.
7. The structure of claim 1, wherein the front surface of the first inner spacer comprises a flat portion and a curved portion.
8. A structure, comprising: a first channel region on a substrate; a second channel region on the first

channel region; a source/drain (S/D) structure adjacent to the first and second channel regions; a first inner spacer between the first and second channel regions and in contact with the S/D structure; a second inner spacer in contact with the first inner spacer; and a third inner spacer between the second inner spacer and the S/D structure.

9. The structure of claim 8, wherein oxide concentrations of the first and second inner spacers are different.

10. The structure of claim 8, further comprises a fourth inner spacer, wherein the third and fourth inner spacers are in contact with opposing side surfaces of the first inner spacer.

11. The structure of claim 10, wherein the third and fourth inner spacers are in contact with the S/D structure.

12. The structure of claim 8, wherein a thickness of the third inner spacer is less than a thickness of the first inner spacer.

13. The structure of claim 8, wherein an interface between the second and third inner spacers is perpendicular to a side surface of the first inner spacer.

14. The structure of claim 8, further comprising a fourth inner spacer, wherein the third and fourth inner spacers are in contact with a curved surface of the first inner spacer.

15. A structure, comprising: a first nano-sheet layer on a substrate; a second nano-sheet layer on a substrate; a first inner spacer between the first and second nano-sheet layers; a second inner spacer on a side surface of the first inner spacer and in contact with the first nano-sheet layer; and a third inner spacer on the side surface of the first inner spacer and in contact with the second nano-sheet layer.

16. The structure of claim 15, wherein the second and third inner spacers are symmetrical with respect to a plane bisecting the first inner spacer.

17. The structure of claim 15, wherein the second inner spacer is isolated from the second nano-sheet layer, and the third inner spacer is isolated from the first nano-sheet layer.

18. The structure of claim 15, wherein the side surface of the inner spacer is curved.

19. The structure of claim 15, further comprising a gate structure between the first and second nano-sheet layers and in contact with the first, second, and third inner spacers.

20. The structure of claim 19, wherein an interface between the gate structure and the first inner spacer is surrounded by the side surface of the first inner spacer.
